

You're provided with two tables: the `advertiser` table contains information about advertisers and their respective payment status, and the `daily_pay` table contains the current payment information for advertisers, and it only includes advertisers who have made payments.

Write a query to update the payment status of Facebook advertisers based on the information in the `daily_pay` table. The output should include the user ID and their current payment status, sorted by the user id.

The payment status of advertisers can be classified into the following categories:

- New: Advertisers who are newly registered and have made their first payment.
- Existing: Advertisers who have made payments in the past and have recently made a current payment.
- Churn: Advertisers who have made payments in the past but have not made any recent payment.
- Resurrect: Advertisers who have not made a recent payment but may have made a previous payment and have made a payment again recently.

Before proceeding with the question, it is important to understand the possible transitions in the advertiser's status based on the payment status. The following table provides a summary of these transitions:

#	Current Status	Updated Status	Payment on Day T
1	NEW	EXISTING	Paid
2	NEW	CHURN	Not paid
3	EXISTING	EXISTING	Paid
4	EXISTING	CHURN	Not paid
5	CHURN	RESURRECT	Paid
6	CHURN	CHURN	Not paid
7	RESURRECT	EXISTING	Paid
8	RESURRECT	CHURN	Not paid

- "Current Status" column: Represents the advertiser's current status.

- "Payment Status" column: Represents the updated payment status based on the conditions
- "Payment on Day T" column: Indicates whether the advertiser made a payment on a specific day (T) or not.

The transitions between payment statuses in the provided table can be summarized as follows:

- Rows 2, 4, 6, and 8: If an advertiser does not make a payment on day T, regardless of their previous status, their payment status transitions to "CHURN" as the updated status.
- Rows 1, 3, 5, and 7: If an advertiser makes a payment on day T, the status is updated to either "EXISTING" or "RESURRECT" based on their previous status. If the previous status was "CHURN," the updated status is "RESURRECT." For any other previous status, the updated status is "EXISTING."

`advertiser` Table:

Column Name	Type
<code>user_id</code>	string
<code>status</code>	string

`advertiser` Example Input:

<code>user_id</code>	<code>status</code>
bing	NEW
yahoo	NEW
alibaba	EXISTING

`daily_pay` Table:

Column Name	Type
<code>user_id</code>	string

paid	decimal
------	---------

daily_pay Example Input:

user_id	paid
yahoo	45.00
alibaba	100.00
target	13.00

Example Output:

user_id	new_status
bing	CHURN
yahoo	EXISTING
alibaba	EXISTING

```
select
COALESCE(a.user_id,d.user_id) AS user_id,
case
when paid is NULL then 'CHURN'
when paid > 0 AND status IN('NEW','EXISTING','RESURRECT') then 'EXISTING'
when paid > 0 AND status IN('CHURN') THEN 'RESURRECT'
when paid is NOT NULL AND a.status IS NULL THEN 'NEW'
else 'NULL' end as new_status
from advertiser a
FULL OUTER JOIN daily_pay d
on a.user_id = d.user_id
ORDER BY user_id;
```